No-arbitrage for lift lending

Bounded TAO-cycles, μ's F, and where the proof fails

Subtensor lending (PR #3268)
Working note — increment 2 (h-sweep + protections)

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Abstract

Following μ's method: for each non-trivial action cycle that **starts and ends holding only TAO**, write the terminal TAO as a factor of the initial holding, sized as a fraction h of pool TAO, as a function of LP state (y, x, w_1) and lending parameters. Then minimise / maximise that factor over the parameter space and ask whether any region has expected value > 1 (attacker profit).

This increment records: (i) state and action set; (ii) μ's September F formula (transcription checked against the hand derivation); (iii) a provisional map of (f_h, φ, w_2) into the current lift design; (iv) the first TAO-pure cycles under forced-escrow longs/shorts; (v) numeric F on SN29-like reserves. Later increments add remaining cycles, closed-form extrema, the weight-ratchet counterexample, and a bounded search that mirrors pallet math.

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1 Method (from μ)

1. Enumerate every non-trivial finite sequence of stake/unstake swaps and lending calls that **begins and ends with a pure-TAO book** (no open position, no residual alpha inventory attributed to the cycle).

2. Write terminal TAO V' as
$$V' = F_{\text{cyc}}(h;\, y, x, w_1;\, \Theta)\, V,$$
where V is the initial TAO stake of the cycle, $h = V/y$ (or another natural scale against pool TAO y), and Θ collects lending parameters (LTV, thin LTV, swap fee, EMA band, APR, term, capacity floors).

3. Over a compact parameter domain (including SN29-like and deep equal-weight pools), compute $\min F_{\text{cyc}}$ and $\max F_{\text{cyc}}$.

4. Positive EV for that cycle $\iff$ some feasible point has $F_{\text{cyc}} > 1$ after fees (attacker convention below).

Remark 1 (Sign convention). We take $F_{\text{cyc}} > 1$ as **attacker profit** (ends with more TAO than started, marking-to-market any returned alpha at the exit swap). μ's September note used "profitable when $F < 1$": that F is a **cost / survival factor for the pool (or for the round-trip price)**, i.e. the reciprocal convention. We keep μ's symbol F for the September formula unchanged, and write F_{cyc} for attacker-return factors. Relation when both apply to the same cycle: $F_{\text{cyc}} = 1/F$ up to fee placement.

2 State and parameters

2.1 LP / Balancer state

- Reserves: TAO y, alpha x.
- Weights: w_1 (base / alpha), $w_2 = 1 - w_1$ (quote / TAO). Spot $p = (w_1/w_2)\,(y/x)$.
- Invariant (Balancer): $L = x^{w_1} y^{w_2}$. At $w_1 = w_2 = \frac{1}{2}$, $L = \sqrt{xy}$ ("$\sqrt{k}$").
- Swap fee: protocol default $\phi_{\text{swap}} = 33/65535 \approx 5.03 \cdot 10^{-4}$. In μ's formula the fee symbol is φ; see §3.1.

2.2 Lending book (per subnet)

Lift book $(L_S, L_L, E_S, B_L, \ldots)$, positions keyed by $(\text{coldkey}, \text{netuid})$, EMA reference for the open band, G_{ema} for capacity. One-year term and 100% APR burn-all interest are part of Θ but are **idle** inside a same-block or few-block cycle (interest ≈ 0).

2.3 Lending parameters Θ (defaults)

Symbol	Default
λ_S (short LTV)	0.25
λ_L (long LTV)	0.25 (0.15 thin)
ϕ_{swap}	33/65535
β (EMA band)	0.15
floor / thin floor / ceiling	0.33 / 0.15 / 0.50
side split	50/50
APR / term	100% / 1 year

3 μ's September F (earlier proposal)

Transcription checked against the hand-derived image:

$$F = (f_h + 1)\left(\frac{(1-\varphi)^{1/w_2}}{(1-\varphi)^{1/w_2} + f_h - \varphi}\right)^{E}, \tag{1}$$

with exponent

$$E = \frac{f_h + 1}{(f_h + 1 + \varphi)\left(1 - \dfrac{1-(1-\varphi)^{1/w_2}}{f_h - \varphi + 1}\right)^{w_2/(1-w_2)}}. \tag{2}$$

Conjecture 1 (μ, earlier design)**.** *Along the September shorting cycle, the pool (or round-trip) factor is F above, and the cycle is attacker-profitable precisely when $F < 1$.*

3.1 Provisional variable map into current lift math

The September proposal predated forced-escrow longs and pull-on-borrow lifts. A working identification used in this note (to be stress-tested against μ's full derivation when available):

Symbol	September reading	Current lift reading
w_2	quote weight	Balancer quote weight w_q (TAO leg)
φ	swap fee taken on the trade	ϕ_{swap} (same order)
f_h	haircut / size as fraction of pool TAO	candidate: $h = P/y$ after lift, or LTV-scaled $h = \lambda \cdot C/y$

Remark 2. If f_h is the **principal as a fraction of post-lift (or pre-trade) pool TAO**, then for a capacity-bound open $h \lesssim 0.075$–0.165 on a side (thin vs floor $\times$ split). For LTV-bound opens with collateral $C = \alpha y$, $P = \lambda C$ gives $f_h = \lambda\alpha$.

4 Does the same cycle exist today?

Cycle 1 (September-style short round trip)**.** Open a forced short of principal P alpha (lift, sell P into the pool, escrow E_S), then close by buying P back (escrow path) or by repaying P alpha from wallet and taking E_S.

Under the **current** design:

- **Sell-path / escrow close** reverses the forced sale on the *lifted* pool then re-adds (τ_L, P) with a weight update that is nearly proportional. This is the intended fee-lossy round trip; $F_{\text{cyc}} < 1$ (attacker) for $\phi > 0$.
- **Wallet-path close** returns $E_S + C$ for alpha repaid from outside, then re-adds (τ_L, P). If that re-add *updates Balancer weights to preserve the post-sale spot*, the September fee analysis is incomplete: an extra **weight-ratchet** term appears (see §6).

So the September cycle *still exists*, but the current implementation adds a weight-update branch that μ's fee-only F does not capture. That is the first place a proof from $F \geq 1$ alone fails.

5 TAO-pure cycles under the current design

Notation: attacker starts with TAO budget $V = h\,y$ (and, where needed, an alpha inventory acquired inside the cycle by a swap that is part of the cycle accounting).

5.1 Cycle A — long wallet ratchet (primary drain candidate)

1. (Optional) Sell alpha inventory into the pool (dump), moving spot down while the lending EMA lags.
2. Open long: post collateral C alpha, lift $g = P/y$, forced-buy escrow A with P TAO, subject to LTV, capacity, EMA band $p' \leq \text{EMA}\,(1+\beta)$.
3. Close with `repay_from_wallet` = `true`: pay P TAO, receive $A + C$, re-add (P, α_L).
4. Sell returned escrow A for TAO.

Mark-to-market: all alpha legs netted; terminal asset is TAO. We write $F_A(h; y, x, w_1; \Theta)$ for V'/V.

Closed form (fee-free sketch, equal weights). Let $h = P/y \in (0,1)$, $w_1 = w_2 = \frac{1}{2}$, ignore swap fees and EMA. Lift removes (P, hx) from (y, x) in the sense $\alpha_L = hx$ when $g = P/y = h$. Forced buy of P on the lifted pool yields

$$A = (1-h)x\Big(1 - \sqrt{\frac{1-h}{1-h+h}}\Big) = (1-h)x\Big(1 - \sqrt{1-h}\Big).$$

Post-open reserves $(y,\ x - \alpha_L - A) = (y,\ x(1-h)\sqrt{1-h})$. Wallet repay + weight update $\to$ reserves $(y+P,\ x-A)$ with spot locked at the post-open price; selling A then gives

$$F_A^{\text{update}}(h) = \frac{\Delta y_{\text{sell}}(A;\ y+P,\ x-A,\ w')}{P},$$

where w' is the reweighted pair. With *fixed* weights,

$$F_A^{\text{keep}}(h) = \frac{y+P}{P}\left(1-\left(\frac{x-A}{x-A+A}\right)^{w_1/w_2}\right) = \frac{1+h}{h}\left(1-\left(1-\frac{A}{x}\right)\right) \quad (\text{for } w_1 = w_2)$$

reduces to a fee-free identity $F_A^{\text{keep}} < 1$ once the forced-buy A is substituted (exact algebra in increment 2). Numerics already show the sign split (§7).

5.2 Cycle B — short wallet mirror

Open short to the EMA band floor, wallet-repay with owned alpha, rebuy alpha (or equivalently value the free E_S against replacing P). Factor F_B.

5.3 Cycle C — raise then wallet-close

`open_long_and_sell` with $S = A$ (near price-neutral by design), then wallet-close the long and sell residual. Factor F_C.

5.4 Cycle D — escrow closes only

Open long or short, close on the escrow AMM path (no wallet repay). Fee-lossy by construction if re-add is the inverse of lift+trade. Factors $F_D^{\text{L}}, F_D^{\text{S}}$.

5.5 Cycle E — partial repay interleaved with swaps

Slice $k \in (0, 1]$, swap, repay remainder. Factor F_E.

6 Where a fee-only proof fails

Proposition 1 (Weight update on asymmetric re-add)**.** *After a forced long buy the pool is at* $(y,\, x - \alpha_L - A)$ *with* ***unchanged*** *weights and elevated spot. Wallet repay credits* (P, α_L) *yielding reserves* $(y + P,\, x - A)$*. Calling* `update_weights_for_added_liquidity` *sets new weights so that the* elevated *spot is preserved. Then* $L = x^{w_1} y^{w_2}$ *(and* w_1*) move monotonically with repeated cycles — matching the observed "*w_1 *and* $\sqrt{k}$ *taking off".*

Consequently μ's F (fees only, fixed weights) is a **lower bound on pool protection** only for cycles that never hit the weight-updating wallet re-add. For Cycle A with weight update,

$$F_A \text{ can exceed } 1$$

even when the September F satisfies $F \geq 1$.

Remark 3 (Fix direction, not yet the proof). Re-adding wallet-repay legs *without* updating weights restores the market-buy residual $(+P, -A)$ under fixed (w_1, w_2). Numerics (§7) show that change flips Cycle A from $F_A > 1$ to $F_A < 1$ on SN29-like pools. A full proof that $F_A \leq 1$ under preserve-weights is deferred to a later increment.

7 Numerics: SN29-like and μ's F

Pool sketch: $y = 10^5$ TAO, $p \approx 0.00279$, $w_1 = w_2 = \frac{1}{2}$, $\varphi = 33/65535$. Side capacity $\sim 0.075\, y$–$0.165\, y$.

7.1 μ's F vs size

Evaluating (1)–(2) at $w_2 = \frac{1}{2}$, $\varphi = 33/65535$ (`scripts/no_arbitrage_F.py`):

f_h	F (fee only)	$F < 1$? (profit)
0.01	1.000485	no
0.05	1.000409	no
0.075	1.000365	no
0.10	1.000323	no
0.165	1.000224	no
0.25	1.000112	no

So on production fees, μ's fee-only F stays *just above* 1 for all listed sizes: the September fee argument alone would *rule out* attacker profit. The drain therefore cannot be explained by fees; it requires an extra mechanism (the weight ratchet).

7.2 Cycle A with weight update (counterexample)

Same pool ($y = 10^5$ TAO, $p \approx 0.00279$), $P = 0.165\,y$, multiplicative swap tax 0.5% per leg in the sketch model:

dump	weights	F_A	profit ()	w_1 after
0%	update	1.0176	+291	0.515
0%	keep	0.9631	−609	0.500
20%	update	1.0302	+498	0.522
20%	keep	0.9513	−804	0.500
35%	update	1.0413	+681	0.528
35%	keep	0.9411	−972	0.500
50%	update	1.0539	+889	0.535
50%	keep	0.9297	−1160	0.500

Counterexample: with weight-updating wallet re-add, $F_A > 1$ on every row; w_1 and L rise. With weight-preserving re-add, $F_A < 1$ on every row. JSON: `scripts/no_arbitrage_F.json`.

8 Theorem status (increment 1)

Conjecture 2 (No TAO-pure arb, preserve-weights wallet re-add)**.** *If wallet-repay re-adds lift legs without updating Balancer weights, and escrow closes re-add after reversing the forced trade, then for every TAO-pure cycle of length $\leq N$ (fees on, EMA fixed or free within band), $F_{\text{cyc}} \leq 1$, with equality only in the zero-fee, zero-size limit.*

Theorem 1 (Status)**.**

1. *Under **current** wallet re-add (weight-updating), Conjecture 2 is **false**: Cycle A is a counterexample with $F_A > 1$ and unbounded w_1/L under repetition while the EMA lags (SN29-like numbers in §7).*
2. *μ's fee-only $F \geq 1$ does **not** imply $F_{\text{cyc}} \leq 1$ for Cycle A, because F omits the weight ratchet.*
3. *Proof of Conjecture 2 under preserve-weights: **open** (this note).*

9 h-sweep and μ's returns graph

9.1 μ's graph (September), eyeball check

Image: return (%) vs pool size in units of the trader's TAO budget (log x, 10%–1000%). Eyeball (verified against the figure):

- $\approx +1\%$ at 10% (pool $= 0.1\times$ budget $\Rightarrow h =$ budget/pool $= 10$ in that proposal's units);
- peak $\approx +4.5\%$ near 100%;
- crosses 0 near $\sim 300\%$;
- $\approx -39\%$ at 1000%.

So that sequence is profitable when the budget is at least about a third of pool TAO — the **large-trade** regime, where lift/AMM curvature dominates fees.

9.2 Cycle A under current lift (production fee)

Sweep $h = P/y$ on an SN29-like pool ($y = 10^5\,\tau$, $p \approx 0.00279$, $\varphi = 33/65535$), weight-update vs preserve-weights (`scripts/no_arbitrage_h_sweep.json`):

h	pool/budget	ret% update	ret% keep	w_1 (upd)
0.01	10000%	−0.09	−0.11	≈ 0.500
0.03	3333%	−0.01	−0.19	≈ 0.500
0.05	2000%	+0.15	−0.35	0.501
0.072	1381%	+0.43	−0.62	0.503
0.165	606%	+2.71	−2.81	0.515
0.30	333%	+9.75	−9.08	0.557
0.50	200%	+31.1	−25.1	0.666

Breakeven for the update path: $h_\star \approx 0.031$ (below: fee-dominated loss; above: ratchet profit). Qualitative agreement with μ: **large h is where EV flips positive**. The quantitative September curve is from the earlier proposal (different fee placement / no forced-escrow lift), so magnitudes differ; both point at the same regime.

10 Do existing protections keep $EV \le 0$?

Reachable h on a **single** open (equal weights):

- EMA band +15%: fee-free buy move $p'/p = (1+h)^{1/w_1} = (1+h)^2 \le 1.15 \Rightarrow h \le \sqrt{1.15} - 1 \approx 0.0724$.
- Side capacity: ≈ 0.075 (thin) to ≈ 0.165 (floor × split).
- `PrincipalExceedsTaoIn` / `AlphaIn`: $P < y$ (and symmetrically for shorts) — far looser than the band/capacity.

At the EMA cap ($h \approx 0.072$) with weight-update Cycle A: $F_A - 1 \approx +0.43\%$ — **still positive EV**. At capacity $h = 0.165$ (only reachable if spot was depressed or the EMA lagged): $\approx +2.7\%$. Preserve-weights is negative at both points.

Splits and multiple coldkeys. Same-side opens are gated by the *sticky* EMA: after one long, spot is up and further longs hit the band sooner. That slows naive splitting, but does not zero EV:

- N sequential opens at $h = 0.05$ (inside the band) each earn $\approx +0.15\%$ while w_1 ratchets ($0.500 \to 0.513$ by $N = 10$).
- A pre-dump that pushes spot below the EMA, then one long up to EMA $\cdot$ 1.15, realises a *much* larger effective h and the 10^2–$10^3\,\tau$ profits of §7.
- Multiple coldkeys in one block still serialize; they share the same post-trade spot/EMA constraint, but can alternate sides or wait for EMA catch-up.

Theorem 2 (Protections vs Cycle A)**.** *EMA band, side capacity, and principal-vs-reserve checks* ***bound*** *h but* ***do not*** *imply $F_A \le 1$ on the weight-updating wallet path. Inside the reachable set, $F_A > 1$ for all $h \gtrsim 0.031$. Only weight-preserving wallet re-add flips the sign through the whole sweep.*

11 Next increments

- Closed forms for F_B, F_C, F_D and grid extrema over (w_1, φ, β).
- Bounded $\le$ 6-step search with runtime differential.
- Call-flow diagram (read state once / validate / derive / act).

References

[1] μ, September 2025 hand derivation of fee factor F (image on PR discussion). Profitability under the earlier proposal when $F < 1$.

[2] *Lift-based lending on main's Balancer pools*, v5.4, `pallets/lending/docs/lift-lending-spec.md`.